

EXPLORATORY DATA ANALYSIS

RAW HOUSE DATA

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Data Summary

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- File 'raw_house_data.csv'
- Contains data on private real-estate that has sold.

MLS	latitude	bedrooms	kitchen_features
sold_price	lot_acres	bathrooms	fireplaces
Zipcode	taxes	sqft_ft	floor_covering
longitude	year_built	garage	HOA

- **Colabs - a cloud-based Jupyter notebook environment**
- **Python libraries:**
 - Pandas – handles data manipulation and analysis
 - Matplotlib: PyPlot – provides MATLAB type plotting interface
 - Seaborn – makes statistical graphics in Python by building on top of Matplotlib and integrates closely with Pandas data structures

Data Cleaning – Nulls and Zeros

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FEATURE	ANALYSIS	ACTION TAKEN
MLS*	0 dups, 0 nulls	No action required.
sold_price	0 nulls	No action required.
zipcode	0 nulls	No action required.
longitude	0 nulls	No action required.
latitude	0 nulls	No action required.
lot_acres	10 nulls, 35 zeros	Nulls replaced by mean. Zeros replaced with mean.
taxes	0 nulls	No action required.
year_built	0 nulls, 5 zeros	Replaced zeros with mean.

MLS* – Multiple Listing Service

Data Cleaning – Nulls and Zeros

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FEATURE	ANALYSIS	ACTION TAKEN
bedrooms	0 nulls, 0 zeros	No action required.
bathrooms	6 nulls, 0 zeros	Nulls replaced with mean.
sqrt_ft	54 nulls	Nulls replaced with mean.
garage	5 nulls	Nulls replaced with 0.
kitchen_ features	33 nulls	Nulls replaced with 'None'.
fireplaces	0 nulls	No action required.
floor_ coverings	1 null	Null replaced with 'None'.

Data Cleaning – Nulls and Zeros

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FEATURE	ANALYSIS	ACTION TAKEN
HOA*	557 nulls	Replaced with -1.

HOA* - Column was HOA fees.

Null means no HOA exist.

Still need to be able to work with that data.

Numbers greater than or equal to 0 were HOA yearly fees.

Data Cleaning – Outliers

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How outliers were determined:

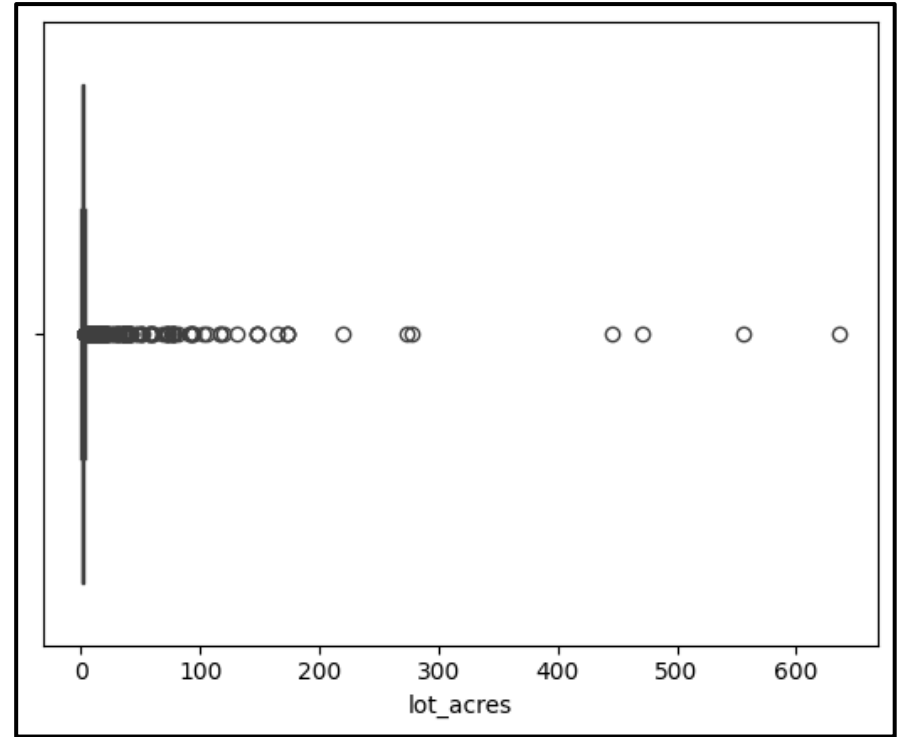
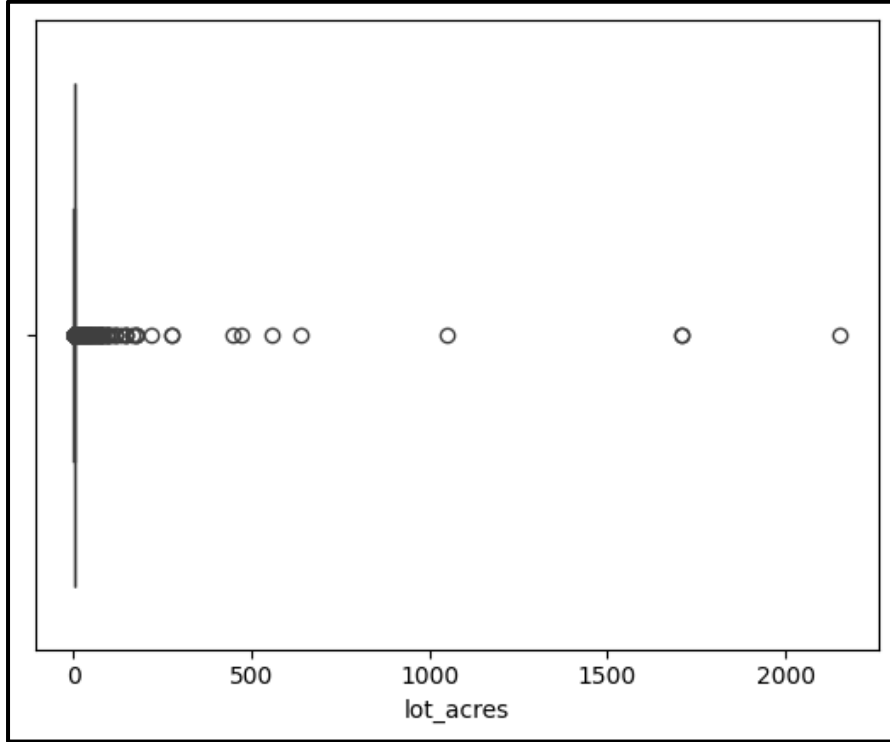
- Viewed boxplots
- Removed those that were 'visibly' far.
- Removed only the bare minimum.

Removed a total of 15 outliers out of 5000 observations.

FEATURE	OUTLIERS REMOVED
lot_acres	4
taxes	2
bedrooms	3
bathrooms	1
HOA	5

Plotting and Graphing

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Data Cleaning – Additional Cleaning Notes

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FEATURE	ACTION TAKEN
MLS	No analysis, this is an indexing feature. Included in cleaned data file.
kitchen_features	Rearranged features list to be alphabetical and no spaces. For example, 'Dishwasher, Freezer, Refrigerator, Oven' became 'Dishwasher,Freezer,Oven,Refrigerator'. Can be used as keys.
fireplaces	Was labeled as 'object' in original data. Now labeled as 'float' in cleaned data.
floor_coverings	Same as with kitchen_features.

The method 'drop_duplicates()' was performed after cleaning.
No duplicates were found.

Saving the Clean Data

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Clean data saved in 'raw_house_data_clean.csv'.

All features were saved to new file cleaned and ready to go.

Plotting and Graphing

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Columns were renamed to be more 'visual-friendly'.

Previous label	New label	Previous label	New label
<code>sold_price</code>	Sold Price	<code>bedrooms</code>	Bedrooms
<code>zipcode</code>	ZIP Code	<code>bathrooms</code>	Bathrooms
<code>longitude</code>	Longitude	<code>sqrt_ft</code>	Sqaure Footage
<code>latitude</code>	Latitude	<code>garage</code>	Garage
<code>lot_acres</code>	Lot Acres	<code>kitchen_features</code>	Kitchen Features
<code>taxes</code>	Taxes	<code>fireplaces</code>	Fireplaces
<code>year_built</code>	Year Built	<code>floor_covering</code>	Floor Covering

Plotting and Graphing

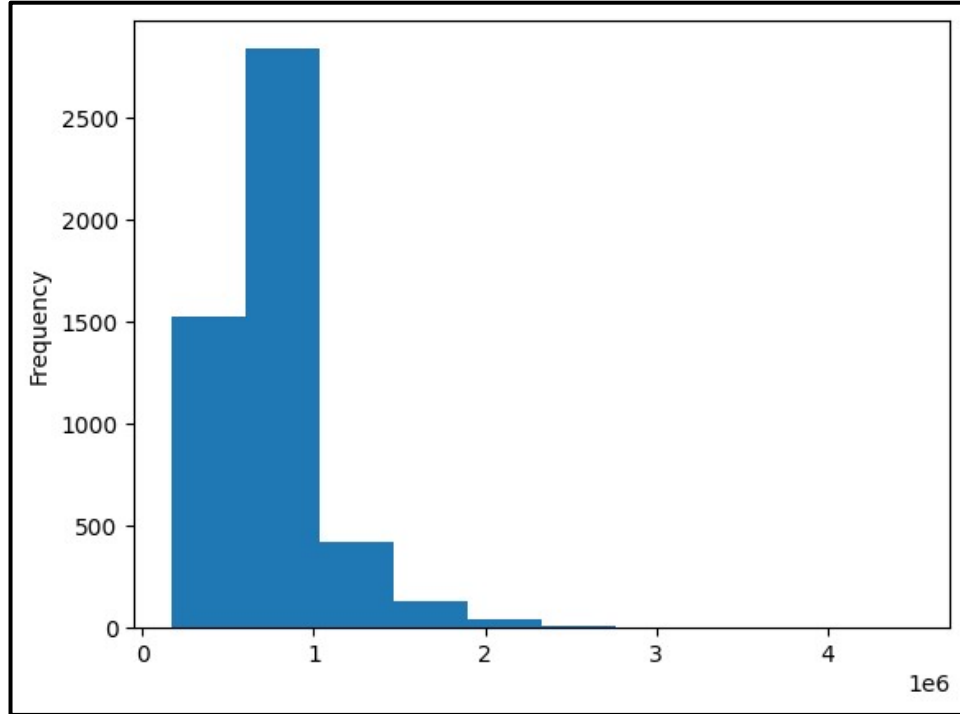
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All features were plotted on HISTOGRAMS, PAIR PLOTS, HEATMAP, and BOXPLOTS *except* those in bold.

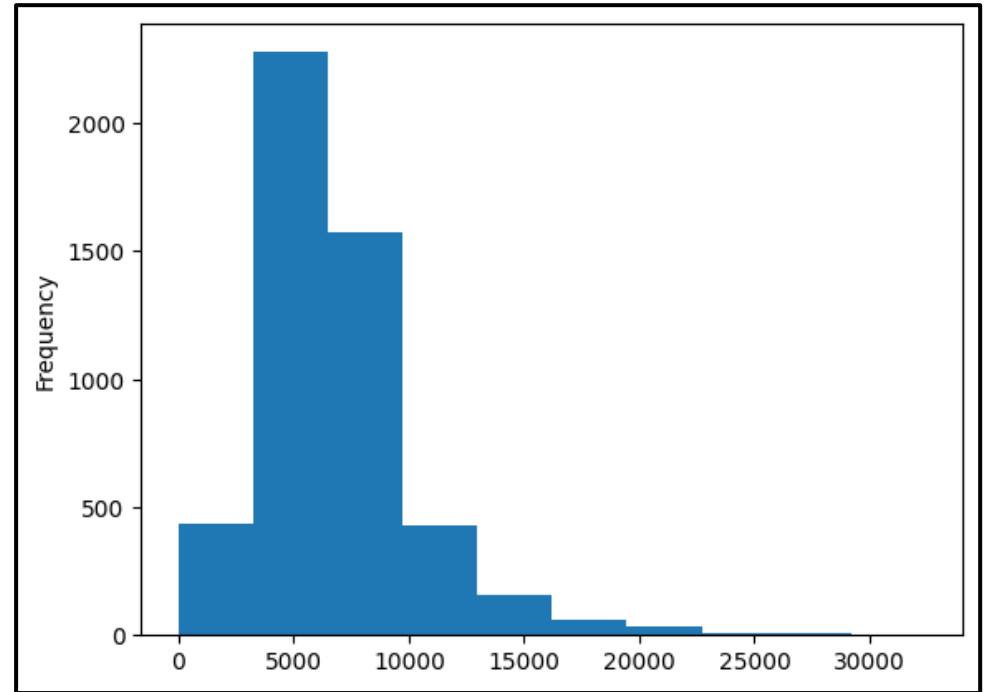
MLS, Sold Price, ZIP Code, Lot Acres, Taxes, Year Built, Bedrooms, Bathrooms, Square Footage, Garage, **Kitchen Features**, Fireplaces, **Floor Covering**, HOA

Plotting and Graphing

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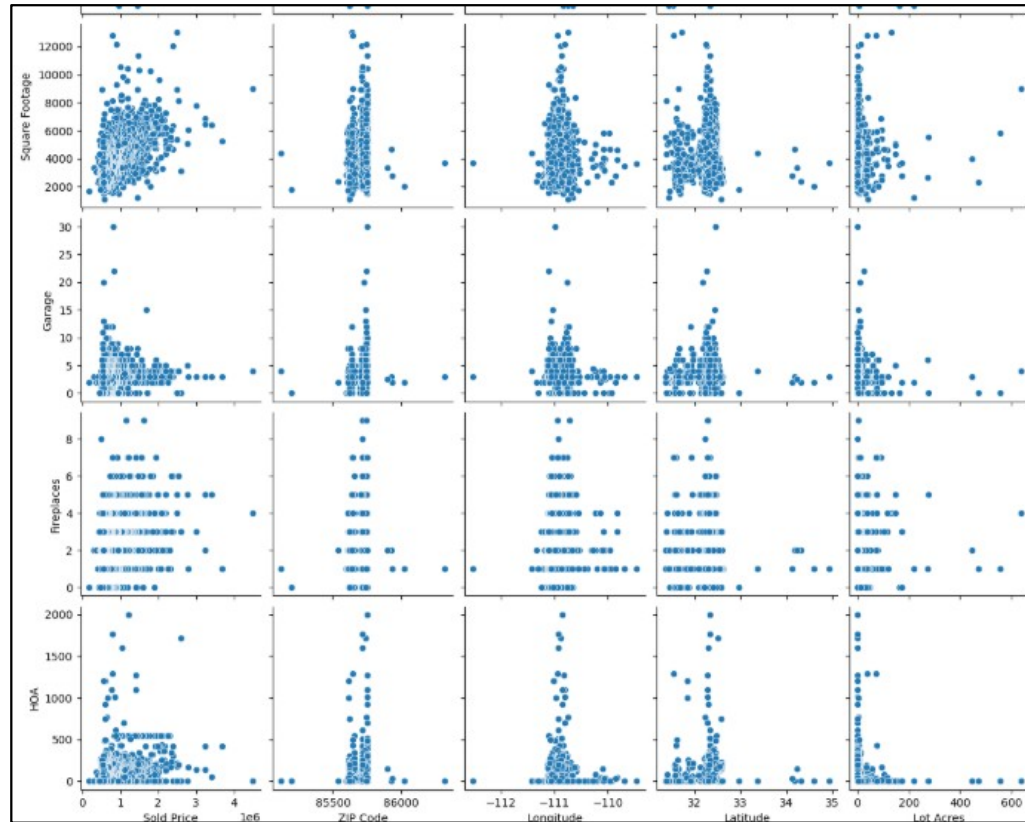
Sold Price



Taxes

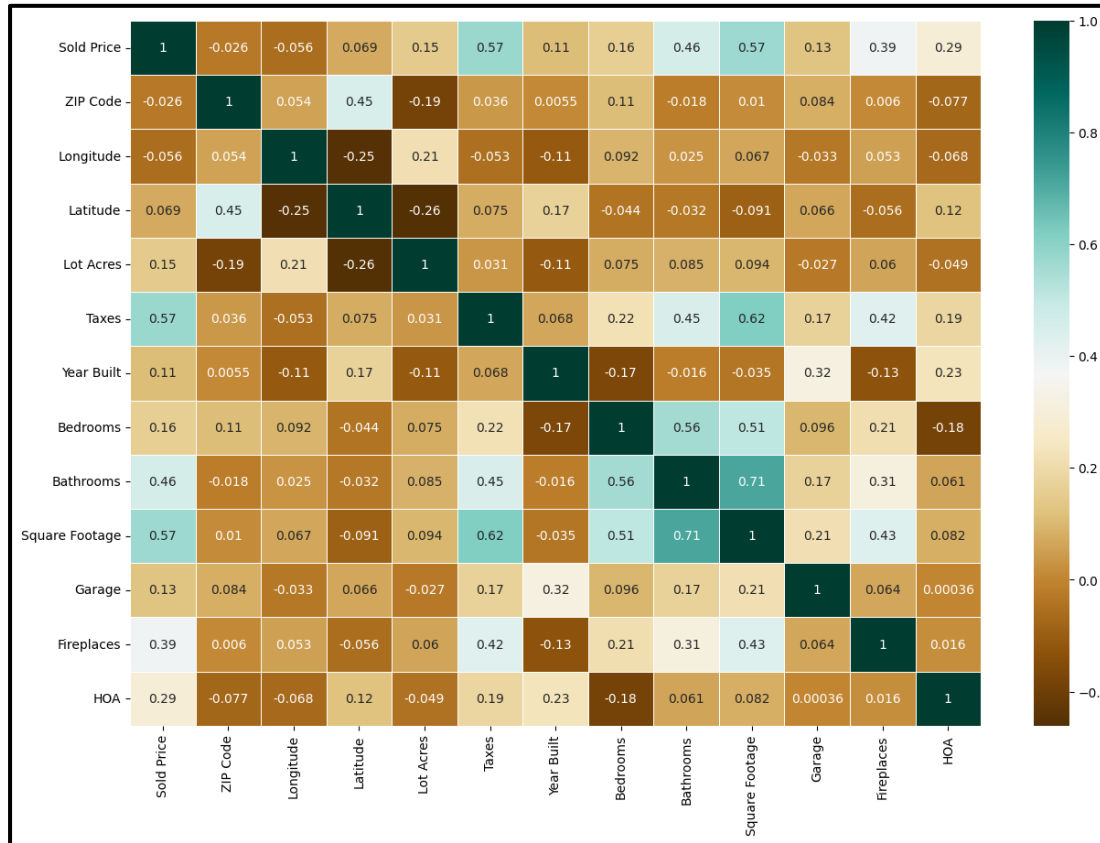
Plotting and Graphing

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Plotting and Graphing

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All data has been cleaned and is ready for further analysis.